

Module 16 Assignment

JavaScript Short Questions

1. Variables & Scoping

What is the difference between **var**, **let**, and **const** in JavaScript?

- **var**: It is function-scoped. A variable declared with **var** can be redeclared and reassigned.
 - **let**: It is block-scoped. A variable declared with **let** can be reassigned but cannot be redeclared within the same scope.
 - **const**: It is also block-scoped. A variable declared with **const** cannot be reassigned after it is declared.
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2. Operators

What is the difference between **==** and **=**? Give one example.

The **==** operator is used to compare two values. JavaScript may convert the data types before comparing them.

The **=** operator is used to assign a value to a variable.

Example:

```
5 == "5" // true
```

Here, **==** compares the values.

```
let x = 5;
```

Here, **=** assigns the value **5** to the variable **x**.

3. Loops

What is the difference between a **for** loop and a **for-of** loop?

A **for** loop is generally used when we need to control the loop using an index or counter.

Example:

```
for (let i = 0; i < 5; i++) {  
  console.log(i);  
}
```

A **for-of loop** is used to directly iterate through the values of an iterable, such as an array or a string.

Example:

```
const numbers = [1, 2, 3];  
  
for (const number of numbers) {  
  console.log(number);  
}
```

4. Functions

What is an arrow function? Write a simple example.

An **arrow function** is a shorter syntax for writing a function in JavaScript. It uses the `=>` symbol.

Example:

```
const add = (a, b) => a + b;
```

This arrow function takes two parameters, `a` and `b`, and returns their sum.

5. Advanced Functions

What are an IIFE, a recursive function, and a generator function? Write one key characteristic of each.

IIFE (Immediately Invoked Function Expression)

An IIFE is a function that runs immediately after it is created.

Key characteristic: It executes immediately and can create a private scope.

Example:

```
(function () {
```

```
    console.log("Hello");
  })();
```

Recursive Function

A recursive function is a function that calls itself.

Key characteristic: It must have a base condition to stop the recursion.

Example:

```
function countdown(n) {
  if (n <= 0) {
    return;
  }

  console.log(n);
  countdown(n - 1);
}
```

Generator Function

A generator function is a special function that can pause and resume its execution.

Key characteristic: It uses the `yield` keyword and is declared using `function*`.

Example:

```
function* generateNumbers() {
  yield 1;
  yield 2;
  yield 3;
}
```

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